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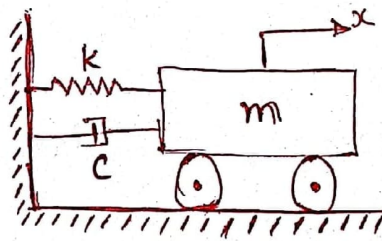
There are mainly four types of damping used in mechanical system.

- i) Viscous damping
- ii) Coulomb damping
- iii) Structural damping
- iv) Non-linear, Slip (or) interfacial damping

Differential equations of damped free vibration

The best example to be considered to do vibration analysis is a mechanical system having spring, mass and damper system as shown in figure below.

It is a well known fact that viscous force is proportional to the velocity across the damper and hence it is written as $c \cdot \frac{dx}{dt} = c\dot{x}$



Free body diagram

Equation of motion is written as

$$m\ddot{x} + c\dot{x} + Kx = 0$$

→ Characteristic equation

Assuming a solution $x = e^{ut}$

where $u \rightarrow$ Constant to be determined.

$$\therefore \dot{x} = ue^{ut} \text{ and } \ddot{x} = u^2e^{ut}$$

$$\therefore \boxed{mu^2e^{ut} + cue^{ut} + ke^{ut} = 0}$$

$$u^2 + \frac{cu}{m} + \frac{K}{m} = 0$$

Solving the above equation for 'u'

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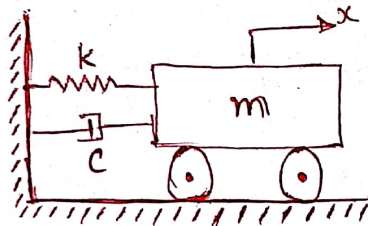
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Solving the above equation for 'u'

$$u = \frac{-c/m \pm \sqrt{c^2/m^2 - 4k/m}}{2}$$

$$\therefore x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\therefore u = \frac{-c}{2m} \pm \sqrt{\left(\frac{c}{2m}\right)^2 - k/m}$$

$\therefore$ the two roots are

$$\begin{aligned} u_1 &= \frac{-c}{2m} + \sqrt{\left(\frac{c}{2m}\right)^2 - k/m} \quad \text{and} \\ u_2 &= \frac{-c}{2m} - \sqrt{\left(\frac{c}{2m}\right)^2 - k/m} \end{aligned}$$

The solution of equation is written as

$$x = A_1 e^{u_1 t} + A_2 e^{u_2 t}$$

where A_1 and A_2 are arbitrary constants

$$\therefore x = A_1 e^{\left[-\frac{c}{2m} + \sqrt{\left(\frac{c}{2m}\right)^2 - k/m}\right]t} + A_2 e^{\left[-\frac{c}{2m} - \sqrt{\left(\frac{c}{2m}\right)^2 - k/m}\right]t}$$

Critical damping Constant and Damping ratio

From the above equation, it is clear that only $\left(\frac{c}{2m}\right)^2 - k/m$ is constant.

Hence critical damping (c_c) is defined as the value of damping coefficient c for which $\left(\frac{c}{2m}\right)^2 - k/m$ is equal to zero.

$$\therefore \left(\frac{c_c}{2m}\right)^2 = \frac{k}{m} \quad \text{ie} \quad \frac{c_c}{2m} = \sqrt{\frac{k}{m}}$$

$$\therefore c_c = 2m \sqrt{k/m} \Rightarrow 2m\omega$$

The ratio c to ' c_c ' is termed as damping ratio (ϵ).

$$\text{ie } \epsilon = \frac{c}{c_c}$$

$$\frac{c}{2m} \text{ can be also written as } \left(\frac{c}{c_c}\right)\left(\frac{c_c}{2m}\right) = \epsilon\omega$$

$$\therefore x_1 = A_1 e^{(-\epsilon + \sqrt{\epsilon^2 - 1})\omega t} + A_2 e^{(-\epsilon - \sqrt{\epsilon^2 - 1})\omega t}$$

ii) Critically Damped System

Based on the value of damping ratio ϵ , the damped systems are put into three categories

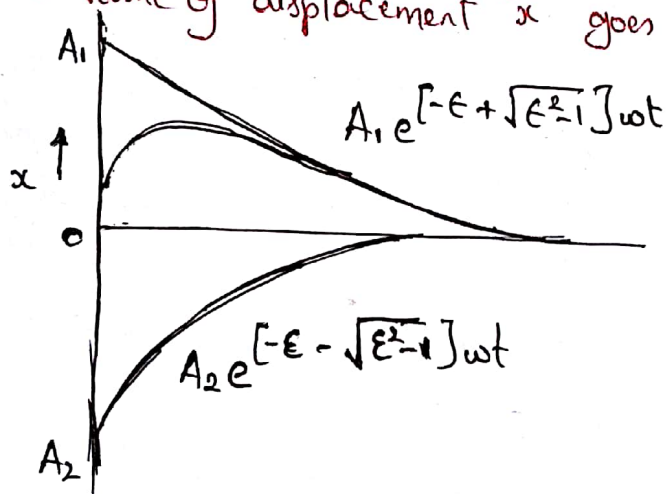
i) Over damped system ($\epsilon > 1$)

When the value of damping ratio is greater than '1', the system is known as over-damped ~~an~~ system.

→ The motion is aperiodic

→ When $t=0$, the displacement is the sum of A_1 and A_2
ie $x = A_1 + A_2$

→ The value of displacement 'x' goes on decreasing with time.



→ The system is non-vibratory in nature

→ The system comes to equilibrium in an exponential manner.

→ Once the system is disturbed, it will take infinite time to come back to equilibrium position.

→ The values of A_1 and A_2 can be determined from initial conditions.

Unit - II - Problems

ii) Critically Damped System

The system is said to be critically damped when $\epsilon = 1$ i.e.

$$\frac{c}{2m} = \sqrt{\frac{k}{m}}$$

The roots of the equation u_1 and u_2 are equal
i.e. $u_1 = u_2 = -\epsilon\omega = -\omega$

So the approximate solution is written as

$$x = A_1 e^{-\omega t} + A_2 t e^{-\omega t} \Rightarrow (A_1 + A_2 t) e^{-\omega t}$$

Applying initial condition to above equation

$$x_0 = A_1$$

$$\dot{x}_0 = \left(\frac{dx}{dt}\right) = A_1(-\omega) + A_2$$

$$\dot{x}_0 = -\omega A_1 + A_2$$

$$\dot{x}_0 = -\omega x_0 + A_2$$

$$A_2 = \dot{x}_0 + \omega x_0$$

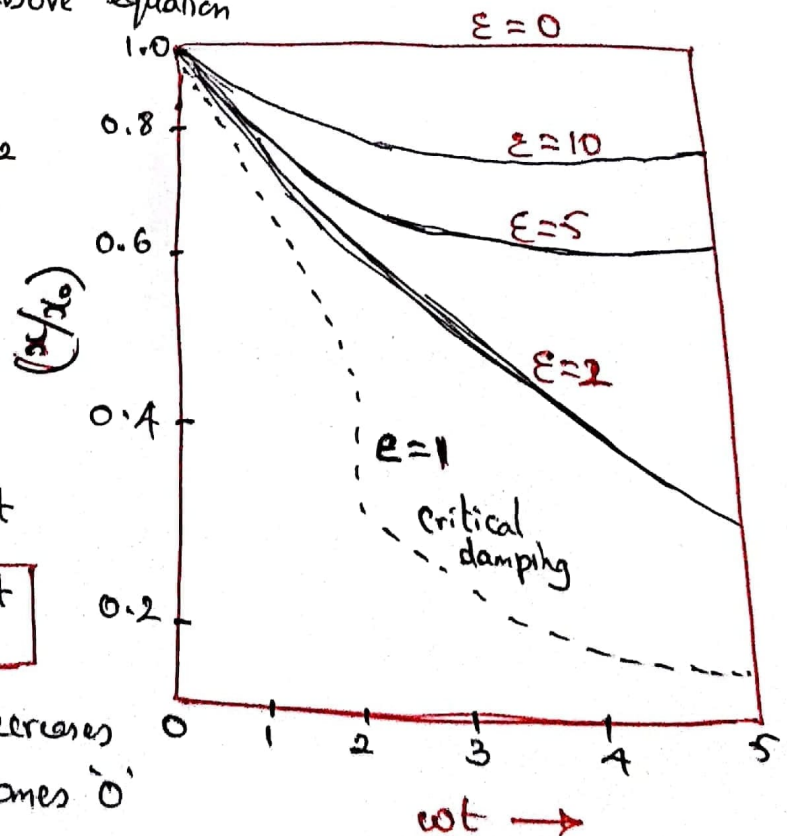
So solution becomes

$$x = x_0 e^{-\omega t} + (\dot{x}_0 + \omega x_0) t e^{-\omega t}$$

$$x = [x_0 + (\dot{x} + \omega x_0)t] e^{-\omega t}$$

We see that the value of $\dot{x}$ decreases as t increases and finally becomes 0 as t tends to infinity.

→ This is also an aperiodic motion



Vibrational Energy and logarithmic decrement

Let us consider at any given instant x_1 be the amplitude and after one cycle, it will be x_2

$$\therefore \delta = \ln \frac{x_1}{x_2}$$

$$\frac{x_1}{x_2} = e^{\delta}$$

$$\frac{x_2}{x_1} = e^{-\delta}$$

On Expanding, we get

$$e^{-\delta} = 1 - \delta + \frac{\delta^2}{2!} - \frac{\delta^3}{3!} + \dots$$

Let E_1 be the vibrational energy at amplitude x_1

$$E_1 = \frac{1}{2} K x_1^2$$

Similarly $E_2 \rightarrow$ Vibrational energy at amplitude x_2 , so

$$E_2 = \frac{1}{2} K x_2^2$$

$$\text{Now } \frac{E_1 - E_2}{E_1} = 1 - \frac{E_2}{E_1} \Rightarrow 1 - \frac{x_2^2}{x_1^2} \Rightarrow 1 - e^{-2\delta}$$

$$= 1 - \left[1 - 2\delta + \frac{(2\delta)^2}{2!} - \frac{(2\delta)^3}{3!} + \dots \right]$$

$$= 2\delta - \frac{(2\delta)^2}{2!} + \frac{(2\delta)^3}{3!}$$

For small δ , higher powers of δ may be neglected.

$$\text{So } \frac{E_1 - E_2}{E_1} = 2\delta$$

$$\boxed{\delta = \frac{\Delta E}{2E_1}}$$

where $\Delta E = E_1 - E_2 =$ Energy dissipated in one cycle.

iii) Under damped System

In underdamped system the value of damping ratio ' ξ ' is less than one. Therefore the roots of the equation can be written as

$$u_1 = [-\xi + j\sqrt{1-\xi^2}]\omega$$

$$u_2 = [-\xi - j\sqrt{1-\xi^2}]\omega$$

where $j = \sqrt{-1} \rightarrow$ Imaginary unit of Complex roots

$\therefore$ the displacement can be written as

$$x = A_1 e^{[-\xi + j\sqrt{1-\xi^2}]\omega t} + A_2 e^{[-\xi - j\sqrt{1-\xi^2}]\omega t}$$

~~$$x = e^{-\xi\omega t}$$~~

$$x = e^{-\xi\omega t} [A_1 e^{j\sqrt{1-\xi^2}\omega t} + A_2 e^{-j\sqrt{1-\xi^2}\omega t}]$$

we know that $e^{jx} = \cos x + j \sin x$

$$\therefore x = e^{-\xi\omega t} [A_1 \cos \sqrt{1-\xi^2}\omega t + A_1 j \sin \sqrt{1-\xi^2}\omega t + A_2 \cos \sqrt{1-\xi^2}\omega t - A_2 j \sin \sqrt{1-\xi^2}\omega t]$$

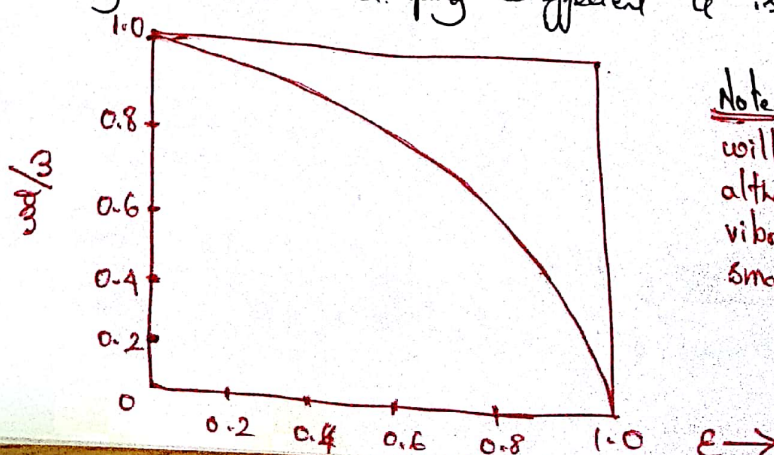
$$x = e^{-\xi\omega t} [C_1 \cos \sqrt{1-\xi^2}\omega t + C_2 \sin \sqrt{1-\xi^2}\omega t]$$

where $C_1 = A_1 + A_2$ and $C_2 = (A_1 - A_2)j$

and $x = C_3 e^{-\xi\omega t} \sin(\sqrt{1-\xi^2}\omega t + \phi_1)$ or $x = C_4 e^{-\xi\omega t} \cos(\sqrt{1-\xi^2}\omega t + \phi_2)$

$\rightarrow$ from the equation, it is clear that, the amplitude decreases exponentially with time

The term $\sqrt{1-\xi^2} \cdot \omega = \omega_d =$ Damped natural frequency
The variation of ω_d with damping Co-efficient ξ is shown below



Note: Theoretically the S/m will never come to rest although the amplitude of vibration may be very very small.

$$\therefore \Rightarrow \frac{\cos(\omega_d t_1 + \phi_2)}{\cos\{(\omega_d t_1 + 2\pi) + \phi_2\}} = 1$$

$$\therefore \frac{x_1}{x_2} = e^{-\varepsilon \omega (t_1 - t_1 - t_d)} \Rightarrow e^{\varepsilon \omega t_d} \Rightarrow e^{\frac{\varepsilon \omega 2\pi}{\omega_d}}$$

$$\frac{x_1}{x_2} = e^{\frac{\varepsilon \omega 2\pi}{\sqrt{1-\varepsilon^2} \omega}} \Rightarrow e^{2\pi \varepsilon / \sqrt{1-\varepsilon^2}}$$

$$\therefore \delta = \log_e \left(\frac{x_1}{x_2} \right) = \frac{2\pi \varepsilon}{\sqrt{1-\varepsilon^2}}$$

When the value of ' ε ' is very small, the above equation is written as

$$\delta = 2\pi \varepsilon$$

If the system executes 'n' cycles, the logarithmic decrement ' δ ' is written as

$$\delta = \frac{1}{n} \log_e \frac{x_1}{x_{n+1}}$$

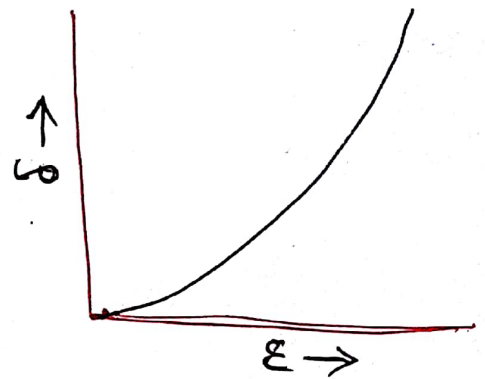
$x_1 \rightarrow$ Amplitude at starting point
 $x_{n+1} \rightarrow$ Amplitude after 'n' cycles.

$$\frac{x_1}{x_2} = \frac{x_2}{x_3} = \frac{x_3}{x_4} = \dots = \frac{x_{n-1}}{x_n} = e^{\delta}$$

$$\left(\frac{x_1}{x_2} \right) \left(\frac{x_2}{x_3} \right) \left(\frac{x_3}{x_4} \right) \dots \left(\frac{x_n}{x_{n+1}} \right) = (e^{\delta})^n$$

$$\therefore n\delta = \ln \frac{x_1}{x_{n+1}}$$

$$\delta = \frac{1}{n} \ln \frac{x_1}{x_{n+1}}$$



Unit - I Problems

① A damping force having magnitude $2 \cos(2\pi t - \pi/4)$ N, gives $5 \cos 2\pi t$ m displacement. Calculate

- i) the energy dissipated during first 5-seconds
- ii) Energy dissipated during the first $3/4$ sec.

Solution: From the given data, it is clear that, the displacement leads the force by $\pi/4 = 45^\circ$.

$$F = F_0 \cos(\omega t - \phi) = 2 \cos(2\pi t - \pi/4)$$

$$x = A \cos \omega t = 5 \cos 2\pi t$$

$$\therefore F_0 = 2, \omega = 2\pi \text{ and } \phi = \pi/4$$

$$A = 5$$

Time period of force 'F' and displacement 'x' is

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2\pi} = 1 \text{ sec}$$

i) In 5-seconds, there will be 5-Complete cycles, therefore.

$$\text{Energy dissipated} = n(\pi) F_0 A \sin \frac{\pi}{4}$$

$$= 5(\pi \times 2 \times 5 \times \sin 45^\circ) = \underline{\underline{111.01 \text{ N-m}}}$$

ii) During $3/4$ sec only part of cycle is complete. To find the energy during this period, we have

$$\begin{aligned} \text{Energy dissipation} &= \int_0^t F \cdot \frac{dx}{dt} \cdot dt = \int_0^{3/4} \left[2 \cos\left(2\pi t - \frac{\pi}{4}\right) \frac{d}{dt} 5 \cos 2\pi t \right] dt \\ &\Rightarrow \int_0^{3/4} \left[2 \cos\left(2\pi t - \frac{\pi}{4}\right) (-5 \times 2\pi) \sin 2\pi t \right] dt \Rightarrow \int_0^{3/4} (-20\pi) \cos\left(2\pi t - \frac{\pi}{4}\right) \sin 2\pi t dt \\ &\Rightarrow \int_0^{3/4} (-20\pi) \left[\cos 2\pi t \cdot \cos \frac{\pi}{4} + \sin 2\pi t \sin \frac{\pi}{4} \right] (\sin 2\pi t) dt \\ &\Rightarrow \int_0^{3/4} (-44.406) \left[\frac{1}{2} \sin 4\pi t + \frac{2}{2} \sin^2 2\pi t \right] dt = \underline{\underline{-40.62 \text{ N-m}}} \end{aligned}$$

UNIT- III - Forced Vibration

- ② Find the equation of motion for the system shown in figure, when
 (a) $\varepsilon = 1.0$, (b) $\varepsilon = 0.3$ and (c) $\varepsilon = 2.0$, if the mass m is displaced
 by a distance of 3 cm and released.

a) When $\varepsilon = 1.0$ [Critically damped System]

$$x = (A_1 + A_2 t) e^{-\omega t}$$

Given displacement $\bar{x} = 3 \text{ cm}$ at $t=0$,

$$\therefore 3 = A_1$$

Differentiating eqn (1) w.r.t $\bar{x}$, we get

$$\dot{x} = -\omega A_1 e^{-\omega t} + A_2 e^{-\omega t} - \omega A_2 t e^{-\omega t}$$

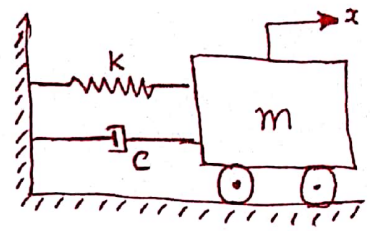
@ $t=0$, we have $\dot{x} = 0$

$$0 = -\omega A_1 + A_2$$

$$\therefore A_2 = +\omega A_1 = 3\omega$$

$\therefore$ therefore displacement $\bar{x}$ is given as.

$$x = (3 + 3\omega t) e^{-\omega t} = 3(1 + \omega t) e^{-\omega t}$$



b) When $\varepsilon = 0.3$ [Under damped s/m]

$$x = A_1 e^{-\varepsilon \omega t} \sin(\sqrt{1-\varepsilon^2} \omega t + \phi) \Rightarrow A_1 e^{-0.3\omega t} \sin(\sqrt{1-0.09} \omega t + \phi)$$

$$x = A_1 e^{-0.3\omega t} \sin(0.953 \omega t + \phi)$$

@ $t=0$, $x=3$

$$\therefore 3 = A_1 \sin \phi$$

Differentiating the displacement equation, we get.

$$\dot{x} = -0.3\omega A_1 e^{-0.3\omega t} \sin(0.953\omega t + \phi) + A_1 e^{-0.3\omega t} (0.953)\omega \cos(0.953\omega t + \phi)$$

@ $t=0$, $\dot{x}=0$

$$\dot{x} = -0.3\omega A_1 e^{-0.3\omega t} \sin(0.953\omega t + \phi) + A_1 e^{-0.3\omega t} (0.953)\omega \cos(0.953\omega t + \phi)$$

@ $t=0$, $\dot{x}=0$

$$\therefore 0 = -0.3\omega A_1 \sin(\phi) + A_1 (0.953\omega) \cos(\phi)$$

$$\therefore A_1 \cos(\phi) \times 0.953 \cancel{\omega} = A_1 \sin(\phi) \times 0.3 \cancel{\omega}$$

$$\tan(\phi) = \frac{0.953}{0.3} = 3.1766$$

$$\therefore \phi = \tan^{-1} 3.1766 = 72^\circ$$

$$\therefore A_1 = 3 / \sin \phi = 3 / \sin(72^\circ) = 3.145$$

$\therefore$ The equation of motion can be written as

$$x = 3.145 e^{-0.3\omega t} \sin(0.953\omega t + 72^\circ)$$

(c) When $\xi = 2.0$ (Over damped η/m)

$$x = A_1 e^{[-\xi + \sqrt{\xi^2 - 1}]\omega t} + A_2 e^{[-\xi - \sqrt{\xi^2 - 1}]\omega t}$$

$$= A_1 e^{(-2 + \sqrt{3})\omega t} + A_2 e^{(-2 - \sqrt{3})\omega t}$$

Applying The initial Conditions, we get

@ $t=0$ ~~$x=3$~~ $x=3 = A_1 + A_2$

ie $A_1 + A_2 = 3$

Differentiating The displacement equation, we get .

~~$$\dot{x} = [-\xi + \sqrt{\xi^2 - 1}]\omega A_1 e^{[-\xi + \sqrt{\xi^2 - 1}]\omega t} + [-\xi - \sqrt{\xi^2 - 1}]\omega A_2 e^{[-\xi - \sqrt{\xi^2 - 1}]\omega t}$$~~

$$\dot{x} = (-2 + \sqrt{3})\omega A_1 e^{(-2 + \sqrt{3})\omega t} + (-2 - \sqrt{3})\omega A_2 e^{(-2 - \sqrt{3})\omega t}$$

@ $t=0$, $\dot{x} = 0$

$$0 = (-2 + \sqrt{3})\omega A_1 + (-2 - \sqrt{3})\omega A_2$$

$$(2 + \sqrt{3})\omega A_2 = (-2 + \sqrt{3})\omega A_1$$

$$A_1 = -13.92 A_2$$

Substituting, we get $A_2 = -0.232$ and $A_1 = 3.232$

$$\therefore x = 3.232 e^{-0.2679\omega t} - 0.232 e^{-3.7322\omega t}$$

- ③ A damper offers resistance 0.05 N at constant velocity 0.04 m/s . The damper is used with $k = 9 \text{ N/m}$. Determine the damping and frequency of the system when the mass of the system is 0.10 kg .

Solution: Damping force $F = c\dot{x}$

$$\text{ie } 0.05 = c \times (0.04)$$

$$\therefore c = 1.25 \text{ N-sec/m}$$

$$\therefore \text{Critical damping} = c_c = 2\sqrt{km} = 2\sqrt{9 \times 0.1}$$

$$c_c = 1.897 \text{ N-sec/m}$$

$$\therefore \text{Damping ratio} = \xi = \frac{c}{c_c} = 0.658$$

The system is underdamped system. The frequency of damped vibration is

$$\omega_d = \sqrt{1 - \xi^2} \cdot \omega = \sqrt{1 - (0.658)^2} \cdot \frac{9}{0.10}$$

$$\boxed{\omega_d = 7.14 \text{ rad/sec}}$$

- ④ A vibratory system in a vehicle is to be designed with the following parameters. $k = 100 \text{ N/m}$, $c = 2 \text{ N-sec/m}$, $m = 1 \text{ kg}$. Calculate the (i) decrease of amplitude from its starting value after 3-Complete oscillations and (ii) the frequency of oscillations.

$$\omega = \sqrt{\frac{k}{m}} = 10 \text{ rad/sec} \quad \text{and} \quad c_c = 2\sqrt{km} = 20 \text{ N-sec/m}$$

$$\xi = \frac{c}{c_c} = \frac{2}{20} = 0.10$$

$$\delta = \frac{2\xi\omega}{\sqrt{1-\xi^2}} = \frac{2 \times 10 \times 0.10}{\sqrt{1-0.01}} = 0.63$$

$$\delta = \ln \frac{x_1}{x_2} \quad \text{and for } n\text{-oscillation} \quad \delta = \frac{1}{n} \ln \frac{x_1}{x_{n+1}}, \quad \text{when } n=3$$

$$\delta = \frac{1}{3} \ln \frac{x_1}{x_{n+1}}$$

$$\therefore \frac{x_1}{x_{n+1}} = e^{(3 \times 0.63)} = e^{1.89} = 6.62 \quad \text{and} \quad \omega_d = \omega \sqrt{1-\xi^2} = 9.95 \text{ rad/sec}$$

⑤ A horizontal spring mass system with coulomb damping has a mass of 5.0 kg attached to a spring of stiffness 980 N/m. If the coefficient of friction is 0.025, calculate

- i) frequency of free oscillations
- ii) the no. of cycles corresponding to 50% reduction in amplitude if the initial amplitude is 5.0 cm and
- iii) the time taken to achieve 50% reduction.

Solution: For a horizontal system, the force is given as

$$F = \mu mg = 0.025 \times 5 \times 9.81 = 1.226 \text{ N}$$

(a) The natural frequency

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{980}{5}} = 14 \text{ rad/sec} = \underline{\underline{2.23 \text{ Hz}}}$$

(b) Amplitude after 50% reduction is half of the initial amplitude
ie $\frac{5 \times 10^{-2}}{2} = 0.025 \text{ m}$

Reduction in amplitude per cycle

$$= \frac{4F}{k} = \frac{4 \times 1.226}{980} = 5 \times 10^{-3} \text{ m}$$

Cycles completed in 50% reduction.

for 1 cycle $\rightarrow 5 \times 10^{-3} \text{ m}$

for 'x' cycles $\rightarrow 0.025$

$$\therefore x = \frac{0.025}{0.005} = 5 \text{ cycles.}$$

(c) Time taken to achieve 50% reduction

$$= \underline{\underline{\text{no. of cycles}}} \cdot \frac{2\pi}{\omega_n}$$

$$= 5 \times \frac{2\pi}{14} = \underline{\underline{2.22 \text{ sec}}}$$

UNIT- III - Forced Vibration

Introduction : There are certain type of vibrations, wherein the amplitude is maintained constant by due to continuous application of external force, which is termed as forced vibrations.

These vibrations are not like the vibrations that can be seen by hitting a Tuning fork once. But can be alike the vibrations due to the same tuning fork, if it is excited with constant loads. Only difference b/w these two is, we can find transient vibrations, if it is hit once and made to vibrate. But the amplitude is maintained almost constant if it is targeted with constant force excitation. The best example to understand this is electric bell.

Hence the type of vibrations which occur under the influence of external force is called external excitation. These excitations may be periodic, impulsive (or) random in nature. And the vibration is called forced vibrations.

- Periodic forces may be harmonic and non-harmonic.
- Vibrations because of impulsive forces are called transient.
- Earthquake is because of random forces.

Equation of motion for forced vibration with Harmonic excitation

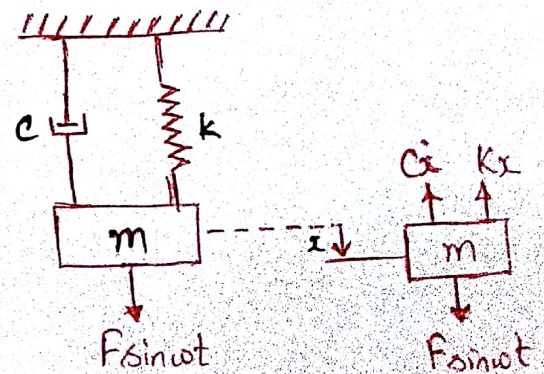
Let us consider a spring mass damper system with external excitation as shown in figure below

Due to the applied load, the mass 'm' is displaced from its equilibrium position by a distance 'x' in downward direction. From the free body diagram, the equation of motion is written as.

$$m\ddot{x} + c\dot{x} + kx = F\sin\omega t$$

This is the second order differential equation with constant coefficients. The general solution of the above equation is of the form

$$x = x_c + x_p$$



$x_c \rightarrow$ Solution of the homogenous equation & called Complementary solution

$x_p \rightarrow$ Particular solution.

from the earlier discussion of free vibration having the equation $m\ddot{x} + c\dot{x} + kx = 0$ (~~homogenous equation~~), we have a solution $x_p = A \sin(\omega t - \phi)$ with respect to harmonic force.

$$\therefore \dot{x}_p = \omega A \cos(\omega t - \phi) \\ = \omega A \sin(\omega t - \phi + \pi/2)$$

$$\ddot{x}_p = \omega^2 A \sin(\omega t - \phi + \pi)$$

Substituting this in the equation of motion, we have.

$$m\omega^2 A \sin(\omega t - \phi + \pi) + c\omega A \sin(\omega t - \phi + \pi/2) + kA \sin(\omega t - \phi) - F \sin \omega t = 0$$

The above equation represents four forces namely

i) Inertia force ii) Damping force iii) Spring force iv) Harmonic force

By applying the above forces, the system is supposed to be in equilibrium

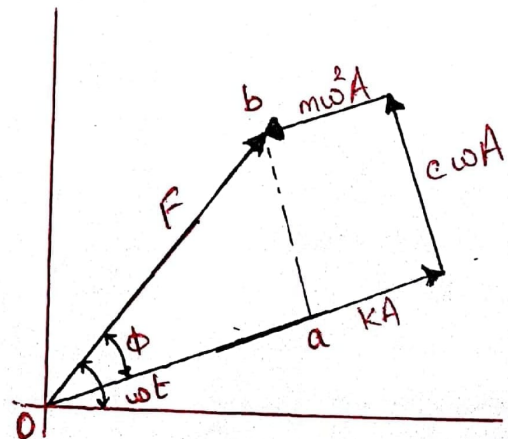
$\Rightarrow$ A vector diagram can be drawn from the above equation, where the spring force is perpendicular to damping force and damping force is perpendicular to inertia force.

from figure, let us consider a triangle 'oab'

$$OB^2 = OA^2 + ab^2$$

$$F^2 = (kA - m\omega^2 A)^2 + (c\omega A)^2$$

$$\therefore A = \frac{F}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$



$$\text{ie } A = \frac{F/k}{\sqrt{\left(\frac{k}{k} - \frac{m\omega^2}{k}\right)^2 + \left(\frac{c\omega}{k}\right)^2}} = \frac{F/k}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left(\frac{c\omega}{k}\right)^2}} \quad \text{and}$$

$$\tan \phi = \frac{ab}{oa} = \frac{c\omega A}{kA - m\omega^2 A} = \frac{c\omega}{k - m\omega^2} = \frac{c\omega/k}{1 - \frac{\omega^2}{\omega_n^2}}$$

$$\text{we know that } \frac{c\omega}{k} = \frac{c}{c_c} \left(\frac{c_c}{2m}\right) \left(\frac{2m}{k}\right) \cdot \omega = \xi \omega_n \frac{2}{\omega_n^2} \cdot \omega \Rightarrow 2\xi \frac{\omega}{\omega_n}$$

Substituting these values in $\tan \phi$, we have

$$\tan \phi = \frac{2\varepsilon \omega/\omega_n}{1 - (\omega/\omega_n)^2} \quad \text{and}$$

$$A = \frac{F/k}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\varepsilon \omega/\omega_n]^2}}$$

Let us take $F/k = X_s$, where $X_s \rightarrow$ Zero frequency deflection

Hence

$$A = \frac{X_s}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\varepsilon \omega/\omega_n]^2}}$$

$$\therefore \frac{A}{X_s} = \frac{1}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\varepsilon \omega/\omega_n]^2}}$$

The non-dimensional quantity (A/X_s) is known as magnification factor or amplitude ratio.

So the particular solution of equation can be written as

$$x_p = X_s \frac{\sin(\omega t - \phi)}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\varepsilon \omega/\omega_n]^2}}$$

The complete solution is written as $x = x_c + x_p$

$$\therefore x = A_1 e^{-\varepsilon \omega_n t} \cos(\sqrt{1 - \varepsilon^2} \omega_n t + \phi_1) + \frac{X_s \sin(\omega t - \phi)}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\varepsilon \omega/\omega_n]^2}}$$

- $\rightarrow$ The values of A_1 and ϕ_1 can be determined from initial conditions
- $\rightarrow$ The frequency and maximum amplitude can be obtained by differentiating the equation with respect to (ω/ω_n) and equating it to zero

$$\text{ie } \frac{d(A/X_s)}{d(\omega/\omega_n)} = \frac{2[1 - (\omega/\omega_n)^2][-2(\omega/\omega_n)] + 2[2\varepsilon \omega/\omega_n][2\varepsilon]}{2\{[1 - (\omega/\omega_n)^2]^2 + [2\varepsilon \omega/\omega_n]^2\}^{3/2}} = 0$$

which leads to $\boxed{\frac{\omega_{\max}}{\omega_n} = \sqrt{1 - 2\varepsilon^2}}$

$\omega_{\max} \rightarrow$ frequency corresponding to maximum amplitude.

At resonance $\omega = \omega_n$, putting this value in magnification factor equation,

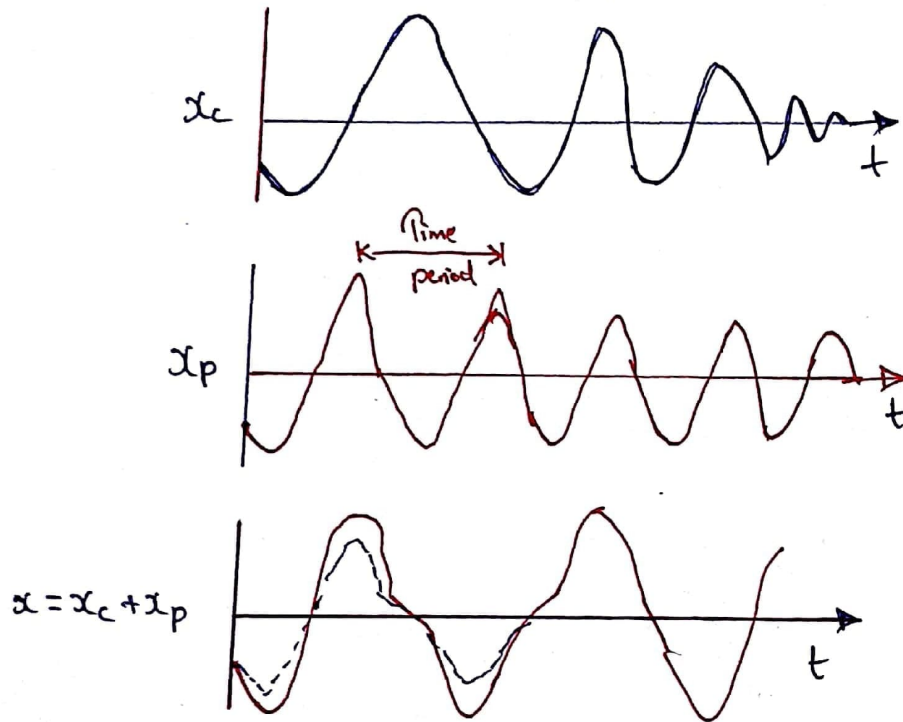
$$\boxed{\frac{A}{X_s} = \frac{1}{2\varepsilon}}$$

Total Response

From the general solution for forced vibration, we know that it is the combination of both Complementary and particular solution.

If we consider the combined effect, the first part of the equation vanishes because of transient vibration, whereas second part remains into existence.

Because of this the amplitude remains constant due to second part and it is called steady vibration.



Characteristic Curves

→ The ratio (ω/ω_n) is called frequency ratio, where ω is frequency of excitation. Similarly (A/x_s) is known as Magnification factor. (or) amplitude ratio.

⇒ The curve between frequency ratio and magnification factor is known as frequency response curve.

⇒ The curve between phase angle and frequency ratio is known as phase-frequency response curve.

$$c\omega x = F_0 \sin \alpha = 0$$

$$c\omega x - m\omega^2 r \sin \alpha = 0 \quad \dots(4.15.4)$$

Substituting the values of $r \cos \alpha$ and $r \sin \alpha$ from equations (4.15.2) and (4.15.1) into equations (4.15.3) and (4.15.4), we get

$$k_1 x - m\omega^2 (x + e \cos \phi) = 0$$

$$(k_1 - m\omega^2) x = m\omega^2 e \cos \phi \quad \dots(4.15.5)$$

$$c\omega x = m\omega^2 e \sin \phi \quad \dots(4.15.6)$$

With the help of above equations (4.15.5) and (4.15.6), we can find the phase angle as

$$\tan \phi = \frac{c\omega x}{(k_1 - m\omega^2) x} = \frac{c\omega}{k_1 - m\omega^2} = \left[\frac{2\epsilon(\omega/\omega_n)}{1 - (\omega/\omega_n)^2} \right] \quad \dots(4.15.7)$$

$$\text{Noting that damping ratio } \epsilon = \frac{C}{2\sqrt{k_1 m}} = \frac{C}{2\sqrt{\frac{k_1}{m}}}$$

$$\epsilon = \frac{C}{2\omega_n m}, \quad 2\epsilon \omega \omega_n = \frac{c\omega}{m}$$

as

$$\omega_n = \sqrt{\frac{k_1}{m}}$$

Squaring and adding equations (4.15.5) and (4.15.6), we get

$$[(k_1 - m\omega^2) x]^2 + [c\omega x]^2 = [m\omega^2 e]^2$$

$$x^2 [(k_1 - m\omega^2)^2 + (c\omega)^2] = [m\omega^2 e]^2$$

$$\frac{x^2}{e^2} = \frac{[m\omega^2]^2}{[(k_1 - m\omega^2)^2 + (c\omega)^2]} = \frac{[\omega^2]^2}{\left[\frac{k_1}{m} - \omega^2 \right]^2 + \left[\frac{c\omega}{m} \right]^2}$$

So

$$\frac{x}{e} = \frac{\omega^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + \left(\frac{c\omega}{m} \right)^2}}$$

$$= \frac{\omega^2}{\sqrt{(\omega_n^2 - \omega^2)^2 + (2\epsilon\omega\omega_n)^2}}$$

$$= \frac{(\omega/\omega_n)^2}{\sqrt{(1 - \omega/\omega_n)^2 + (2\epsilon\omega\omega_n/\omega_n^2)^2}} \quad (\text{dividing by } \omega_n^2)$$

$$\frac{x}{e} = \frac{\beta^2}{\sqrt{(1 - \beta^2)^2 + (2\epsilon\beta)^2}} \quad \dots(4.15.8)$$

where $\beta = \omega/\omega_n$

SOLVED EXAMPLES

EXAMPLE 4.1. A vibrating system having mass 1 kg is suspended by a spring of stiffness 1000 N/m and it is put to harmonic excitation of 10 N. Assuming viscous damping, determine

- the resonant frequency
- the phase angle at resonance
- the amplitude at resonance
- the frequency corresponding to the peak amplitude and
- damped frequency

Take $C = 40 \text{ N-sec/m}$

SOLUTION. (a) Frequency at resonance

$$\omega = \omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{1000}{1}} = 31.62 \text{ rad/sec}$$

Damping factor ϵ is given as

$$\epsilon = \frac{c}{2m\omega_n} = \frac{40}{2 \times 1 \times 31.62} = 0.632$$

(b) Phase angle at resonance

$$\tan \phi = \infty$$

So $\phi = \pi/2$

(c) Amplitude at resonance

$$x_{\text{res}} = \frac{F}{c\omega_n} = \frac{10}{40 \times 31.62} = 7.9 \times 10^{-3} \text{ m}$$

(d) The frequency corresponding to the peak amplitude

$$\omega_{\text{peak}} = \omega_n \sqrt{1 - 2\epsilon^2} = 31.62 \sqrt{1 - 2(0.632)^2} = 14.14 \text{ rad/sec}$$

(e) The damped frequency is given as

$$\omega_d = \sqrt{1 - \epsilon^2} \omega_n = \sqrt{1 - (0.632)^2} \times 31.62 = 24.5 \text{ rad/sec}$$

EXAMPLE 4.2. The vibrating system as shown in figure 4.6 is displayed for vibrational characteristics. The total mass of the system is 25 kg. At speed of 1000 rpm, the system and the eccentric mass have a phase difference of 90° and the corresponding amplitude is 1.5 cm. The eccentric unbalanced mass of 1 kg has a radius of rotation 4 cm. Determine (a) the natural frequency of the system (b) the damping factor (c) the amplitude at 1500 rpm, and (d) the phase angle at 1500 rpm.

SOLUTION. Given $m = 25 \text{ kg}$

$A = 1.5 \text{ cm}$

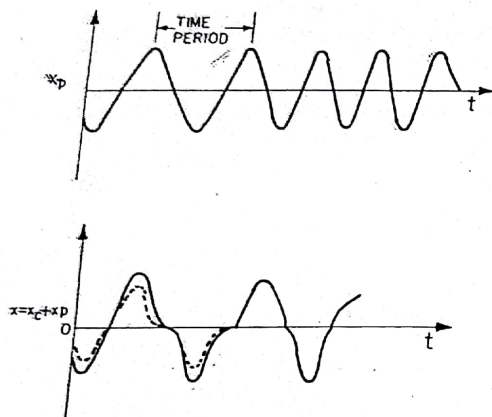


Fig. 4.3.

4.3.2. Characteristic Curves

The ratio ω/ω_n is called the frequency ratio where ω is the frequency of excitation. Similarly, A/X_s is known as magnification factor or amplitude ratio.

A curve between frequency ratio and magnification factor is known as frequency response curve. Similarly a curve between phase angle and frequency ratio is known as phase-frequency response curve. Both curves which are drawn with the help of equations (4.3.10) and (4.3.8) as shown in figure 4.4.

The following points are noted from these equations and figures :

1. At zero frequency magnification is unity and damping does not have any effect on it.
2. Damping reduces the magnification factor for all values of frequency.
3. The maximum value of amplitude occurs a little towards left of resonant frequency.
4. At resonant frequency the phase angle is 90° .
5. The phase angle increases for decreasing value of damping above resonance.
6. The amplitude of vibration is infinite at resonant frequency and zero damping factor.

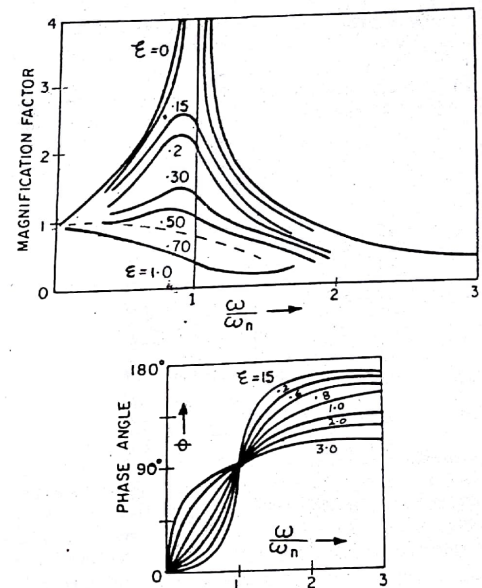


Fig. 4.4.

7. The amplitude ratio is below unity for all values of damping which are more than 0.70.
8. The variation in phase angle is because of damping. Without damping it is either 180° or 0° .

4.3.3. Variation of Frequency Ratio ω/ω_n

There are three possibilities of ω variation i.e., $\omega < \omega_n$, $\omega = \omega_n$ and $\omega > \omega_n$. This variation of ω will affect the magnitude of various forces acting on the system. The three cases are discussed here.

(a) $\omega/\omega_n \ll 1$

We know that inertia and damping forces are given by the expressions $m\omega^2 A$ and $c\omega A$ respectively. It is clear that both the terms are affected by ω . When the value of ω is very small, inertia and damping forces are reduced considerably resulting the small value of ϕ in figure 4.2. Hence, the phase

angle is reduced and F is balanced by spring force kA which are almost equal and opposite in magnitude. See figure 4.5 (a).

(b) $\omega/\omega_n = 1$

When the frequency of excitation ω increases and becomes equal to the natural frequency ω_n , resonance occurs. The phase angle becomes 90° . Inertia force and spring force are found to be equal and opposite. Excitation force balances the damping force i.e. $c\omega A = F$. Thus giving the amplitude of vibration at resonance

$$A = F/c\omega \quad \dots(4.3.3.1.)$$

See figure 4.5(b).

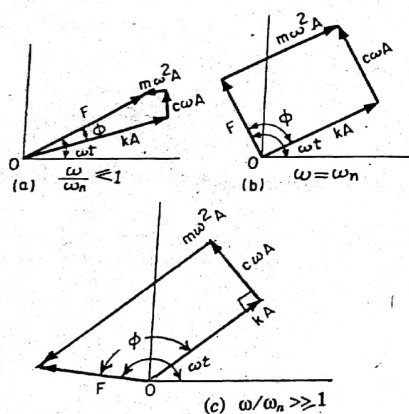


Fig. 4.5.

(c) $\omega/\omega_n \gg 1$

At very high frequencies ω , inertia force increases very rapidly and its magnitude is very large. Damping and spring force are small in magnitude. When the value of ω/ω_n is very high, the phase angle ϕ is very close to 180° . See figure 4.5 (c).

4.4. RESPONSE OF A ROTATING AND RECIPROCATING UNBALANCE SYSTEM

A machine having rotor as one of its components is called a rotating machine, like turbines and I.C. engines. The problem of unbalance in such a system occurs when centre of gravity of rotor does not

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coincide with the axis of rotation. The distance between axis of rotation and the centre of gravity is called eccentricity e and the mass acting at a distance e from the axis of rotation is known as eccentric mass m_0 . A rotating machine supported on springs is shown in figure 4.6. The mass of the machine including the eccentric mass is m and the angular speed of rotor as ω rad/sec.

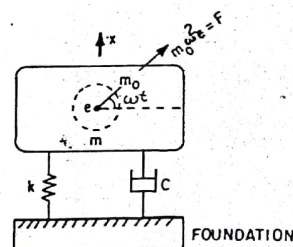


Fig. 4.6. Rotating unbalance.

k and c being the spring stiffness and damping coefficient respectively. The system may be assumed to have single degree of freedom as it can move in one direction only. Let the eccentric mass m_0 make an angle ωt with the reference axis. At any moment vertical displacement is given as $x + e \sin \omega t$. The centrifugal force of eccentric mass will be $m_0 \omega^2 e$. This force can be resolved into two components, i.e., vertical and horizontal. Vertical component is of much significance which acts as equivalent excitation and can be expressed as $m_0 e \omega^2 \sin \omega t$.

The differential equation of motion can be written as

$$(m - m_0) \frac{d^2 x}{dt^2} + m_0 \frac{d^2}{dt^2} (x + e \sin \omega t) + kx + c \frac{dx}{dt} = 0$$

$$(m - m_0) \ddot{x} + m_0 \ddot{x} - m_0 \omega^2 e \sin \omega t + kx + c \dot{x} = 0$$

or

$$m \ddot{x} + c \dot{x} + kx = m_0 \omega^2 e \sin \omega t \quad \dots(4.4.1)$$

This equation represents damped forced vibration. Comparing this equation with equation (4.3.1), we see that $m_0 \omega^2 e = F$ and rest of the things are same. $m_0 e \omega^2$ acts as the excitation force on the system.

To find the amplitude A and phase angle ϕ , above equation can be solved like equation (4.3.1).

$$A = \frac{m_0 e \omega^2 / k}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [(2c\omega/\omega_n)^2]}}$$

$$\frac{A}{\frac{m_0 e}{m}} = \frac{(\omega/\omega_n)^2}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\epsilon \omega/\omega_n]^2}} \quad \left(\because \omega_n^2 = \frac{k}{m} \right) \quad \dots(4.4.2)$$

At resonance $\omega = \omega_n$

$$\frac{A}{\frac{m_0 e}{m}} = \frac{1}{2\epsilon} \quad \dots(4.4.3)$$

The phase angle can be written as

$$\tan \phi = \frac{2\epsilon(\omega/\omega_n)}{1 - (\omega/\omega_n)^2} \quad \dots(4.4.4)$$

The complete solution in this case can be written as

$$\begin{aligned} x &= x_c + x_p \\ &= A_1 e^{-\epsilon \omega_n t} \cos(\sqrt{1 - \epsilon^2} \omega_n t + \phi_2) \\ &\quad + \frac{m_0 e \omega^2 / k}{\sqrt{[1 - (\frac{\omega}{\omega_n})^2]^2 + [2\epsilon \frac{\omega}{\omega_n}]^2}} \end{aligned} \quad \dots(4.4.5)$$

The variation of $A \frac{m_0 e}{m}$ with frequency ratio for various values of damping factor is shown in figure 4.7 while the curve between the phase angle and frequency ratio is the same as shown in figure 4.4. The following points are concluded from this equation and figure :

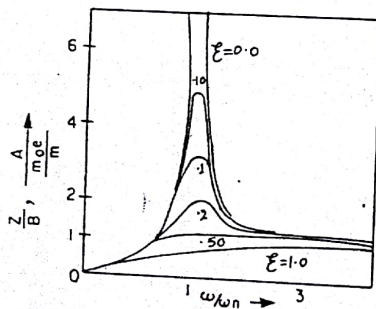


Fig. 4.7. Amplitude frequency response curve.

1. Damping factor plays very important role during resonance in controlling the amplitudes. It reduces the amplitude A .

2. When the value of ω is very small as compared to ω_n , it is known as low speed system. For a low speed system the value of $\left(\frac{A}{\frac{m_0 e}{m}}\right) \rightarrow 0$.
3. Similarly, for a high speed system ω is very high, then $\left(\frac{A}{\frac{m_0 e}{m}}\right) \rightarrow 1$.
4. At very high speed the effect of damping seems to be negligible.
5. A comparison between figures 4.4 and 4.7 shows that peak amplitude occurs to the right of resonance in figure 4.7 and on the contrary the peak amplitude falls to the left of resonance in figure 4.4.
6. At resonance $\omega = \omega_n$ and so

$$\frac{A}{\frac{m_0 e}{m}} = \frac{1}{2\epsilon}$$

$$A_{\text{resonance}} = \frac{m_0 e}{2m\epsilon}$$

Considering a reciprocating engine as shown in figure 4.7 (a), let the equivalent mass of the reciprocating part be m_0 and the total mass

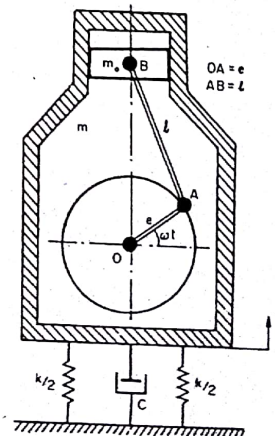


Fig. 4.7 (a). Reciprocating Unbalance.

of the engine including the reciprocating part is m . The crank length and the connecting rod length are e and l respectively.

Inertia force due to reciprocating mass m_0 is approximately equal to $F = m_0 e \omega^2 \left[\sin \omega t + \left(\frac{e}{l} \right) \sin 2\omega t \right]$. If e is very small as compared to l , second harmonic of the above equation i.e. $\frac{e}{l} \sin 2\omega t$ part can be neglected and the exciting force F becomes equal to $m_0 e \omega^2 \sin \omega t$ which is the same as that for rotating unbalance. Hence, for small value of e , the previous analysis of rotating unbalance is applicable to this also.

4.5. SUPPORT MOTION

In case of locomotives or vehicles the wheels act as base or support for the system. The wheels can move vertically up and down on the road surface during the motion of the vehicle.

At the same time there is relative motion between the wheels and the chassis. So chassis is having motion relative to the wheels and the wheels are having motion relative to the road surface. The amplitude of vibration in case of support motion depends on the speed of vehicle and nature of road surface. The vibration measuring instruments are designed on the support motion approach. Such systems are supposed to have single degree of freedom for the simplicity of mathematical expression. In a vibratory system where the support is put to excitation absolute and relative motion become important from subject point of view.

4.5.1. Absolute Motion

Absolute motion of a mass means its motion with respect to the coordinate system attached to the earth. As shown in figure 4.8, the absolute displacement of support is $y = B \sin \omega t$ and the absolute dis-

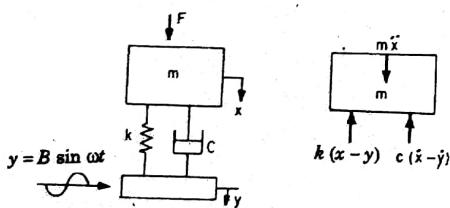


Fig. 4.8.

placement of the mass m from its equilibrium position is x . The displacement of mass m relative to the support is z . The net elongation of the spring is $(x - y)$ and the relative motion between the two ends of the damper is $(\dot{x} - \dot{y})$. Then $z = x - y$ and $\dot{z} = \dot{x} - \dot{y}$.

The equation of motion can be written as

$$m\ddot{x} + c(\dot{x} - \dot{y}) + k(x - y) = 0$$

or

$$m\ddot{x} + c\dot{x} + kx = c\dot{y} + ky \quad \dots(4.5.1.1)$$

The support is subjected to harmonic vibration, $y = B \sin \omega t$. Substituting this value of y in equation (4.5.1.1), we get

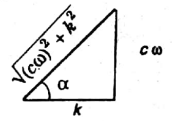
$$\begin{aligned} m\ddot{x} + c\dot{x} + kx &= cB\omega \cos \omega t + kB \sin \omega t \\ &= B \sqrt{k^2 + c^2\omega^2} \left[\frac{k}{\sqrt{k^2 + c^2\omega^2}} \sin \omega t + \frac{c\omega}{\sqrt{k^2 + c^2\omega^2}} \cos \omega t \right] \\ &= B \sqrt{k^2 + c^2\omega^2} [\cos \alpha \sin \omega t + \sin \alpha \cos \omega t] \\ m\ddot{x} + c\dot{x} + kx &= B \sqrt{k^2 + c^2\omega^2} \sin(\omega t + \alpha) \quad \dots(4.5.1.2) \end{aligned}$$

and

$$\tan \alpha = \frac{c\omega}{k}$$

where

$$\alpha = \tan^{-1} \frac{c\omega}{k} = \tan^{-1} (2\epsilon \omega / \omega_n) \quad \dots(4.5.1.3)$$



Steady state solution can be written as

$$x = A \sin(\omega t + \alpha - \phi) \quad \dots(4.5.1.4)$$

Comparing equation (4.5.1.2) with equation (4.3.1), we see that

$$F = B \sqrt{k^2 + c^2\omega^2}$$

Let us go back to equation (4.3.9) i.e.

$$A = \frac{F/k}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [(2\epsilon \omega/\omega_n)^2]}}$$

Steady state amplitude can be written from equation (4.5.1.2) as

$$\begin{aligned} A &= \frac{B \sqrt{k^2 + c^2\omega^2}/k}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [(2\epsilon \omega/\omega_n)^2]}} \\ \frac{A}{B} &= \frac{\sqrt{1 + (2\epsilon \omega/\omega_n)^2}}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [(2\epsilon \omega/\omega_n)^2]}} \quad \dots(4.5.1.5) \end{aligned}$$

$$\frac{F}{k} = \text{static deflection} \left(\therefore \frac{c\omega}{k} = \frac{C}{C_c} \left(\frac{C_c}{2m} \right) \left(\frac{2m}{k} \right) \omega = \epsilon \omega_n \frac{2}{\omega_n^2} \omega = 2\epsilon \frac{\omega}{\omega_n} \right)$$

The ratio $\frac{A}{B}$ is called the displacement transmissibility, which is the ratio of the amplitude of the body to the amplitude of the support.

We know that $\tan \alpha = \frac{c\omega}{k}$

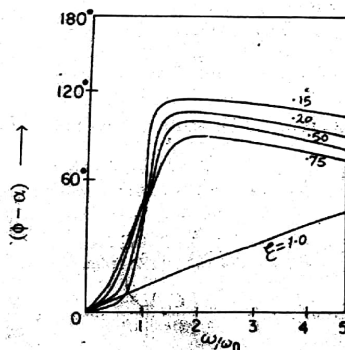
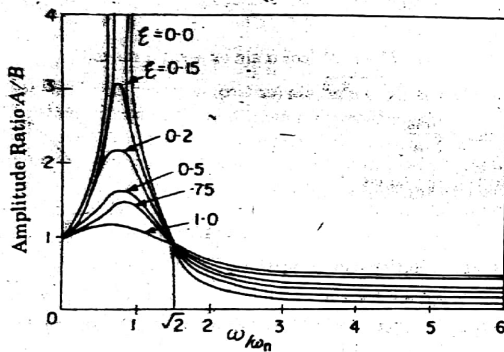
or

$$\alpha = \tan^{-1} \frac{c\omega}{k} = \tan^{-1} 2\xi \omega / \omega_n$$

and

$$\tan \phi = \frac{2\xi \omega / \omega_n}{1 - (\omega / \omega_n)^2}$$

$$\phi = \tan^{-1} \frac{2\xi \omega / \omega_n}{1 - (\omega / \omega_n)^2}$$



So phase $(\phi - \alpha)$ can be written as

$$\phi - \alpha = \tan^{-1} \left[\frac{2\xi \omega / \omega_n}{1 - (\omega / \omega_n)^2} \right] - \tan^{-1} \left(2\xi \frac{\omega}{\omega_n} \right) \quad \dots(4.5.1.6)$$

With the help of equations (4.5.1.5) and (4.5.1.6) steady state amplitude and phase are shown in figure 4.9. We conclude the following points from this figure :

1. At frequency ratio $\omega / \omega_n = \sqrt{2}$ for all values of damping the amplitude ratio (A/B) is unity.
2. The amplitude ratio is less than one for all values of damping when $\omega / \omega_n > \sqrt{2}$.
3. For high values of frequency ratio ($\omega / \omega_n \gg \sqrt{2}$), the amplitude ratio is quite small which means that the support displacement is almost negligible.
4. For high values of frequency ratio the damping does not have any important role so far as the amplitude ratio is concerned.
5. The phase angle is not 90° at resonance as in case of force excitation (see figure 4.4).

4.5.2. Relative Motion

In section (4.5.1), we have assumed that z is the relative displacement of the mass with respect to the support. So it can be written as

$$\begin{aligned} z &= x - y \\ \dot{z} &= \dot{x} - \dot{y} \\ \ddot{z} &= \ddot{x} - \ddot{y} \end{aligned} \quad \dots(4.5.2.1)$$

Substituting the above relations in equation (4.5.1.1), we get

$$\begin{aligned} m(\ddot{z} + \ddot{y}) + c\dot{z} + kz &= 0 \\ m\ddot{z} + c\dot{z} + kz &= -m\ddot{y} \end{aligned}$$

The sinusoidal support excitation is assumed in the system i.e. $y = B \sin \omega t$. Substituting the value of y in the above equation, we have

$$m\ddot{z} + c\dot{z} + kz = m\omega^2 B \sin \omega t \quad \dots(4.5.2.2)$$

This equation is known as relative equation of motion and it is of the same form as equation (4.4.1) i.e.

$$m\ddot{x} + c\dot{x} + kx = m_0 \omega^2 e \sin \omega t$$

Let us assume that the steady state relative amplitude Z lags the excitation by angle α , so

$$z = Z \sin (\omega t - \alpha)$$

$\frac{Z}{B}$ ratio can be written as

$$\frac{Z}{B} = \frac{(\omega/\omega_n)^2}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\epsilon \omega/\omega_n]^2}} \quad \dots(4.5.2.3)$$

and

$$\alpha = \tan^{-1} \left[\frac{2\epsilon \omega/\omega_n}{1 - (\omega/\omega_n)^2} \right] \quad \dots(4.5.2.4)$$

The characteristics of this equation are shown in figures 4.7 and 4.4 for frequency.

The following conclusions are drawn from these figures and equation :

1. When $(\omega/\omega_n) > 3$, the amplitude ratio $\left(\frac{Z}{B}\right)$ is almost unity. It means that the relative amplitude Z and the support amplitude B are equal. This is the principle on the basis of which vibration measuring instruments are designed which will be discussed later on in this chapter.
2. For $\left(\frac{Z}{B}\right)$ ratio being unity, the mass m will be having no displacement.
3. Damping does not have any effect on $\frac{Z}{B}$ ratio for high values of ω/ω_n , i.e. for $\omega/\omega_n > 3$.

4.6. VIBRATION ISOLATION

The high speed engines and machines when mounted on foundations and supports cause vibrations of excessive amplitude because of unbalanced forces set up during their working. These are the disturbing forces which damage the foundation on which the machines are mounted. So the vibrations transmitted to the foundation should be eliminated or reduced considerably by using some devices such as springs, dampers, etc., between the foundation and the machine. These devices isolate the vibrations by absorbing some disturbing energy themselves and allow only a fraction of it to pass through them to the foundation. Thus the amplitude of vibration is minimised and the adjoining structure or foundation is not put to heavy disturbances. The isolation is expressed in terms of force or motion. Lesser the amount of force or motion transmitted to the foundation greater is said to be the isolation. So machines are mounted on isolation mounts. There are two basic requirements for an isolator : Firstly, there should be no rigid connection between the unit (Machine, engine or vibrating body, etc.) and the base otherwise the undesired vibrations will be completely transmitted from the unit to the base. It may damage the supporting structure.

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Secondly, it should be ensured that the isolators remain together in case the damping material (rubber, cork, felt, pad, etc.) fails. It should be just to keep the machine or unit in the safe position with respect to the support.

The materials normally used for vibration isolation are rubber, felt cork, metallic springs, etc. These are put between the foundation and the vibrating body.

Rubber as an isolator is quite useful for shear loading. Its sound transmissibility is very low. Its properties are influenced by heat, gasoline, oil, etc. So it cannot be used at high temperature in the presence of gasoline and oil. It is preferred for light loads and high frequency oscillations.

The damping factor of felt is high so it is specially used for low frequency ratios. Its pads are put between the machine and the support. It is preferred to use many pads at a time rather than a single large pad of the same size.

Cork is suitable for compressive loads. It is not perfectly elastic. At high loads it becomes more flexible.

Metal springs used are of two types namely helical spring and leaf spring. The spring has high sound transmissibility which can be reduced by covering it with pads of felt, cork or rubber. The spring can be used in all working conditions as it is not affected by air, water, oil or usual temperature variation. They are useful for high frequency ratios.

4.7. TRANSMISSIBILITY

Let us consider figure 4.8 for analysis. The spring and damping forces are kx and $c\dot{x}$ respectively.

The amplitude of vibration is A , so the maximum value of these forces will be kA and $c\omega A$ respectively. The forces are perpendicular to each other, because of phase difference being 90° so their resultant F_T is given as

$$F_T = \sqrt{(kA)^2 + (c\omega A)^2} = A\sqrt{k^2 + c^2\omega^2} \quad \dots(4.7.1)$$

In the above equation F_T is the force transmitted to the foundation. The disturbing force is F . The ratio of F_T to F is called transmissibility. So it can be expressed mathematically, as

$$T.R. = \frac{F_T}{F} \quad \dots(4.7.2)$$

From equation (4.2.9), we have

$$F = Ak \sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\epsilon \omega/\omega_n]^2} \quad \dots(4.7.3)$$

Substituting the values of F and F_T in equation (4.6.2), we have

$$\begin{aligned} T.R. &= \frac{A\sqrt{k^2 + c^2\omega^2}}{Ak\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\varepsilon\omega/\omega_n]^2}} \\ &= \frac{\sqrt{1 + (c^2/k^2)\omega^2}}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\varepsilon\omega/\omega_n]^2}} \\ T.R. = \frac{F_T}{F} &= \frac{\sqrt{1 + (2\varepsilon\omega/\omega_n)^2}}{\sqrt{[1 - (\omega/\omega_n)^2]^2 + [2\varepsilon\omega/\omega_n]^2}} \quad \dots(4.7.4) \end{aligned}$$

The above equation is the same as equation (4.4.1.5), so

$$\frac{F_T}{F} = \frac{A}{B} \quad \dots(4.7.5)$$

If $\frac{F_T}{F} = 0.25$, then the system is said to have 25% transmissibility.

The following informations are available from equation (4.7.4) and figure 4.9:

1. The area of vibration isolation starts when transmissibility is less than unity and $\omega/\omega_n > \sqrt{2}$. Thus when the frequency ω of exciting force is given, the isolation mounts can be designed such that $\omega/\omega_n > \sqrt{2}$. If ω is small the value of ω_n should be small and the value of natural frequency $\left(\omega_n = \sqrt{\frac{k}{m}}\right)$ results from high value of mass or low value of stiffness. Thus if the disturbing frequency ω is low the isolation can be accomplished by adding certain amount of mass to the system.
2. Damping is important specially at resonance otherwise TR will be excessively large.
3. In the vibration isolation region $\omega/\omega_n > \sqrt{2}$, so equation (4.6.4) in absence of damping can be written as

$$T.R. = \frac{1}{(\omega/\omega_n)^2 - 1}$$
4. Unity value of T.R. occurs at two points where ω/ω_n is zero and $\sqrt{2}$ for all values of damping.
5. Under ideal operating conditions, T.R. must be zero. Hence, ω/ω_n should be as large as possible. Thus, the natural frequency of the system should be small i.e. weak spring with fairly heavy mass.

4.8 FORCED VIBRATIONS WITH COULOMB DAMPING

(Refer section 3.2 also)

For a single degree of freedom system with coulomb damping or dry friction damping, subjected to a harmonic force $F \sin \omega t$, the equation of motion is given by

$$M\ddot{x} + kx \pm \mu R_N = F \sin \omega t \quad \dots(4.8.1)$$

An exact solution is available due to early work of J.P. Den Hartog for small damping force so that the motion is continuous. For high value of dry damping force the motion does not remain continuous. If the dry friction force is small compared to harmonic force, an approximate solution is required for this case.

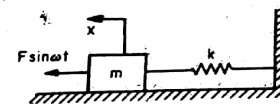


Fig. 4.10.

The sign of frictional force (μR_N) is positive when the mass moves from left to right and negative when the mass moves from right to left. By finding an equivalent viscous damping ratio, we find an approximate solution of eqn. (4.8.1). For viscous damping ratio, we equate the energy dissipated due to dry friction to the energy dissipated by an equivalent viscous damper during a full cycle of motion. The energy dissipated by friction force μR_N in a quarter cycle is $\mu R_N A$. So in a full cycle, the energy dissipated by dry friction damping is given by

$$\Delta E = 4 \mu R_N A = 4 F_f A \quad \dots(4.8.2)$$

where F_f = Frictional force

A = Amplitude of motion

If equivalent viscous damping is denoted by C_e , the energy dissipated during a full cycle is given by eqn. (3.2.5).

$$\Delta E = \pi C_e \omega A^2 \quad \dots(4.8.3)$$

Equating the above two equations, we obtain an equivalent viscous damping factor which is measured in terms of dry friction

$$\pi C_e \omega A^2 = 4 F_f A$$

$$C_e = \frac{4 F_f}{\pi \omega A} = \frac{4 \mu R_N}{\pi \omega A} \quad \dots(4.8.4)$$

The steady state amplitude for the system having viscous damping is given by eqn. (4.3.6).

$$A = F/k / \sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left(\frac{C\omega}{k}\right)^2}$$